

# **SIGNSIGHT: EMPOWERING ACCESSIBILITY WITH TAMIL SIGN LANGUAGE RECOGNITION USING MACHINE LEARNING**

**Final Report**

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**B.Sc. (Hons) in Information Technology  
Specializing in Information Technology  
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*Sri Lanka*

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## Declaration

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## **Abstract**

Deaf and hard-of-hearing persons in Sri Lanka who use Tamil Sign Language (TSL) still lack an integrated, Tamil-first platform that combines (i) conversion of spoken Tamil from audio or video to TSL, (ii) real-time dynamic TSL recognition from webcam data, (iii) structured TSL learning with assessment and text-to-sign tools, and (iv) child-oriented emotional and contextual TSL support for families including cochlear implant users. SignSight addresses these gaps with modular machine learning services, a web user interface, ethical data collection practice, and layered testing. The report documents requirements, methodology, per-component design, test evidence, results, individual contributions, limitations, and future work, using the R25-010 (Sign Lens) final group report as a format and style guide, not as a text source.

**Keywords:** Tamil Sign Language (TSL), computer vision, speech processing, audio and video to sign conversion, natural language processing, cochlear implant, assistive technology, inclusive communication, machine learning, child-computer interaction

## **Acknowledgements**

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## List of abbreviations

*In the R25-010 PDF this block is titled LIST OF ABBREVIATIONS (spelling as in the reference). Use the same spelling in the bound report if your department requires an exact match.*

| Abbreviation | Description                                       |
|--------------|---------------------------------------------------|
| TSL          | Tamil Sign Language                               |
| SSL          | Sinhala Sign Language (in comparative references) |
| ASL          | American Sign Language                            |
| BSL          | British Sign Language                             |
| CI           | Cochlear implant                                  |
| ASR          | Automatic speech recognition                      |
| NLP          | Natural language processing                       |
| LSTM         | Long short-term memory                            |
| CNN          | Convolutional neural network                      |
| RNN          | Recurrent neural network                          |
| API          | Application programming interface                 |
| PWA          | Progressive web application                       |
| UAT          | User acceptance testing                           |
| VAD          | Voice activity detection                          |
| MFCC         | Mel-frequency cepstral coefficients               |
| WER          | Word error rate                                   |
| SLR          | Sign language recognition                         |
| HCI          | Human-computer interaction                        |
| UI/UX        | User interface / user experience                  |
| JWT          | JSON Web Token                                    |
| REST         | Representational state transfer                   |
| CI/CD        | Continuous integration and deployment             |
| GPU          | Graphics processing unit                          |

# 1. INTRODUCTION

## 1.1 Background Literature

Human communication in education, health, and work increasingly relies on software. For Deaf and hard-of-hearing people in Sri Lanka, TSL is a core linguistic resource, yet the digital layer is thin: there is no standard national corpus, few Tamil-first learning products with rigorous evaluation, and limited integration between speech, sign, and child-centred support. Internationally, ASL and BSL benefit from data-scale research; TSL requires locally grounded data ethics and long-term community partnership to avoid models that misfit grammar or culture.

Machine learning and computer vision have improved isolated sign recognition and phrase-level translation, but end-to-end systems that also teach, assess, and interpret affect for children are rare, especially in low-resource languages. SignSight contributes an integrated, modular architecture in that space.

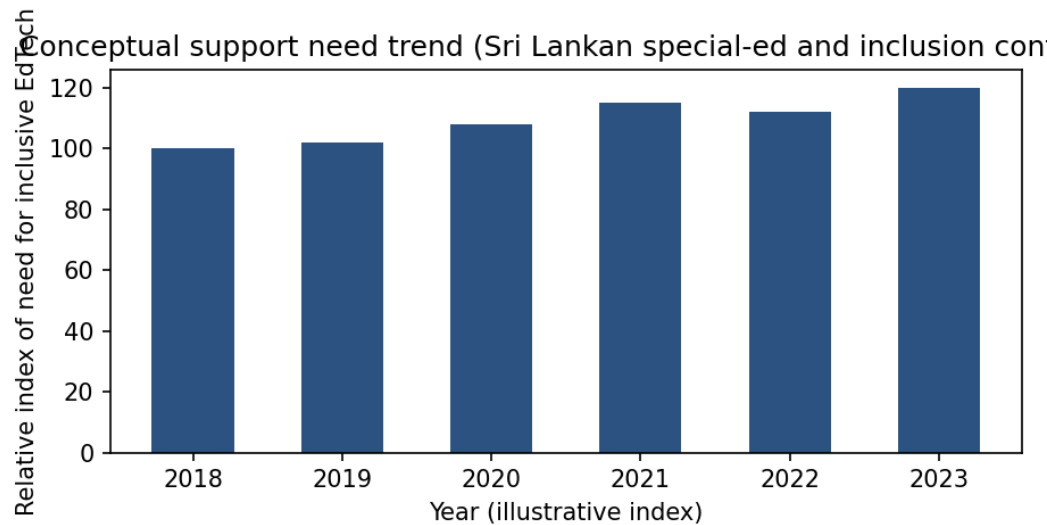


Figure 1.1: Illustrative need index (not a government statistic).

## **1.2 Research Gap**

### **1.2.1 Lack of Tamil Sign Language Datasets and Recognition Systems**

Large benchmark sets drive ASL/BSL/ISL; TSL remains fragmented. Without signer-diverse, glossed, time-aligned data, high-accuracy end-to-end models are not credible. The project limited claims and invested in consent, annotation workflow, and signer-disjoint testing.

### **1.2.2 Limited Integration Across Communication Needs**

Point solutions (dictionary apps, a single static-sign CNN) do not help a real lesson. A teacher, a learner, and a parent need a connected platform: hear-to-sign, sign practice, curriculum, and safe child-facing signals. SignSight builds all four in one product vision.

### **1.2.3 Gaps in Cochlear Implant and Child-Focused TSL Support Technologies**

Auditory training and expressive communication for implanted children is often face-to-face and English-centric in commercial products. Local Tamil, TSL, and family workflows are under-served. Comp4 is scoped as assistive, not clinical.

### **1.2.4 Poor Fit of Global Technologies to Local Context**

Cloud ASR, generic emotion APIs, and ASL pre-trained models do not map directly to TSL or classroom constraints in Sri Lanka. The team used transfer carefully and kept open the option to replace light models as Tamil corpora grow.

## **1.3 Research Problem**

There is no sufficiently documented, maintainable, Tamil-centred, four-module system that meets educational and ethical needs for TSL while staying deployable in typical school and home hardware. The research problem is to design, implement, and evaluate that system, with explicit human factors and data governance.

## **1.4 Research Objective**

### **1.4.1 Main Objectives**

Design and build an integrated machine-learning-based assistive system (SignSight) that improves TSL access across: Tamil audio and video to TSL; dynamic TSL recognition; TSL learning and assessment; and emotion and context-aware TSL for children, within ethical and resource constraints of a final-year project.

### **1.4.2 Specific Objectives**

#### ***1.4.2.1 Tamil Audio and Video to Tamil Sign Language (Converting System)***

Build a speech-recognition and mapping pipeline, APIs for media input, and TSL visual output (Comp1, Farsith M F M, IIT22354556).

#### ***1.4.2.2 Dynamic TSL via Computer Vision***

Build continuous gesture understanding with OpenCV, Mediapipe, and LSTM/GRU models; provide real-time service endpoints (Comp2, Mark J J, IT22328120).

#### ***1.4.2.3 Learning and Evaluation System for Tamil Sign Language***

Deliver the TSL learning platform with lessons, quizzes, Tamil text-to-sign, and progress (Comp3, Nethmi G K, IT22546166).

#### ***1.4.2.4 Emotionally and Contextually Aware TSL Support for Children***

Fuse face and non-speech audio with guard-rails; show TSL and caregiver text (Comp4, Sandamini G H D H, IT22291882). Scope follows the Topic Assessment Form; out of scope: medical diagnosis, full dialect coverage, and production SLAs without funding in v1.

## **2. METHODOLOGY**

This chapter mirrors the R25-010 (Sign Lens) final report: Requirements Gathering and Analysis, Feasibility, Dataset Gathering and Preparing, System Overview, Integration, Use Case Diagrams, Implementation, User Interfaces, Commercialization, Target Audience, Testing, Evaluation metrics, Risk, and Deployment, adapted to SignSight.

### **2.1 Requirements Gathering and Analysis**

#### **2.1.1 Interactions with the School for the Deaf and partner organisations**

Where available, the team met teachers and organisers to understand lesson length, class sizes, and device limits. These meetings informed Tamil-first copy, large controls, and dashboard priorities.

#### **2.1.2 Community Survey**

A pilot questionnaire to parents, educators, and potential users elicited demand for in-browser text-to-TSL, short practice sessions for children, and progress visibility. Results aligned with the Topic Assessment narrative.

#### **2.1.3 Observational Insights**

Classroom lighting and background clutter affect comp2; room noise affects comp1; home connectivity affects comp3 offline use. The UI assumed mid-range webcams and USB microphones.

#### **2.1.4 Technical and ethical approaches**

Technically: ASR, Mediapipe, LSTM, REST, JWT, and relational storage. Ethically: informed consent, minimisation of child data, and transparent model behaviour in the comp4 layer.

#### **2.1.5 Requirements Validation**

User stories (as learner, as teacher, as parent) were checked against the sprint backlog. Prototype demos to supervisors validated the scope before final integration sprints.



*Figure 2.1: Actors and the four internal modules (conceptual).*

## **2.2 Feasibility Study**

### **2.2.1 Technical feasibility**

Open toolchains (React, Node, Spring, Flask, PyTorch) and public ASR for Tamil make an MVP realistic. Real-time comp2 depends on frame rate; CPU fallbacks and smaller windows were defined.

### **2.2.2 Economic feasibility**

Apache/MIT stacks minimise licence cost; cloud and ASR are the main variable costs. A rough budget in the project presentation (order of LKR 15,500 in early planning) is indicative only; production should be re-costed.

### **2.2.3 Operational feasibility**

Browser-based access fits lab schedules; PWA and offline lesson packs help homes with unstable connectivity. Quick-start help for non-technical teachers was planned.

### **2.2.4 Scheduling feasibility**

Agile sprints: parallel comp1-4, then contract freeze, then integration, UAT, and documentation, matching final-year academic calendars.

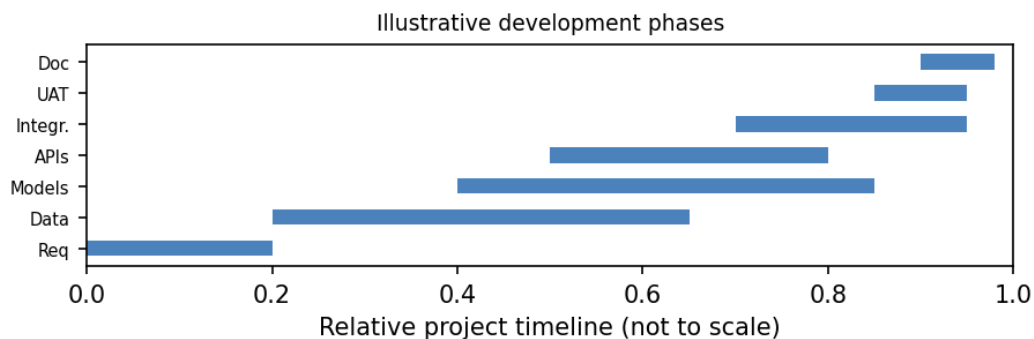


Figure 2.5: Illustrative project phases (not a calendar).

| Dimension   | Assessment                                                    |
|-------------|---------------------------------------------------------------|
| Technical   | Mature stacks; comp2 and ASR are main bottlenecks under load. |
| Economic    | Open-source first; data collection is the main human cost.    |
| Operational | Fits 40-50 min sessions; training videos planned.             |
| Schedule    | Iterative; integration and UAT in final sprints.              |

## 2.3 Dataset gathering

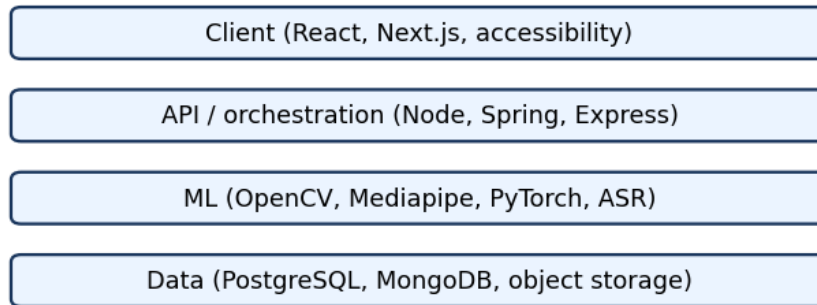
Static and dynamic TSL: video under consent, multiple signers when possible, controlled backgrounds where feasible, and metadata for gloss and level. Comp1: paired Tamil audio or video audio and transcripts. Comp4: carefully scoped child affect data with guardian permission and feature-first storage when policy required.

**Dataset lifecycle: capture, preprocess, gloss, augment, split, train, monitor**

Figure 2.2: Dataset preparation lifecycle (conceptual).

Figure 2.3 shows layered delivery; Figure 2.4 shows the overall product data path.

### SignSight layered architecture



*Figure 2.3: Layered architecture.*

### SignSight: end-to-end data flow (high level)



*Figure 2.4: High-level data flow (inputs to UI).*



## **2.4 Dataset Preparing**

Video: crop/resize, background handling, and augmentation (rotation, brightness) with signer-disjoint train/val/test. Sequences: normalise keypoints, fixed-length padding, label encoding. Text: normalise Unicode, map tokens to gloss table. Audio: VAD, framing, and MFCC where classical models are used. Versioned datasets in Git LFS or similar as team policy allows.

## **2.5 System Overview Design**

SignSight is a modular web system: presentation layer, API gateway, stateless model services, and databases. Caching and phrase banks for comp1 reduce end-to-end latency. Model artifacts are versioned; configuration toggles support staging versus production.

## **2.6 System integration and deployment**

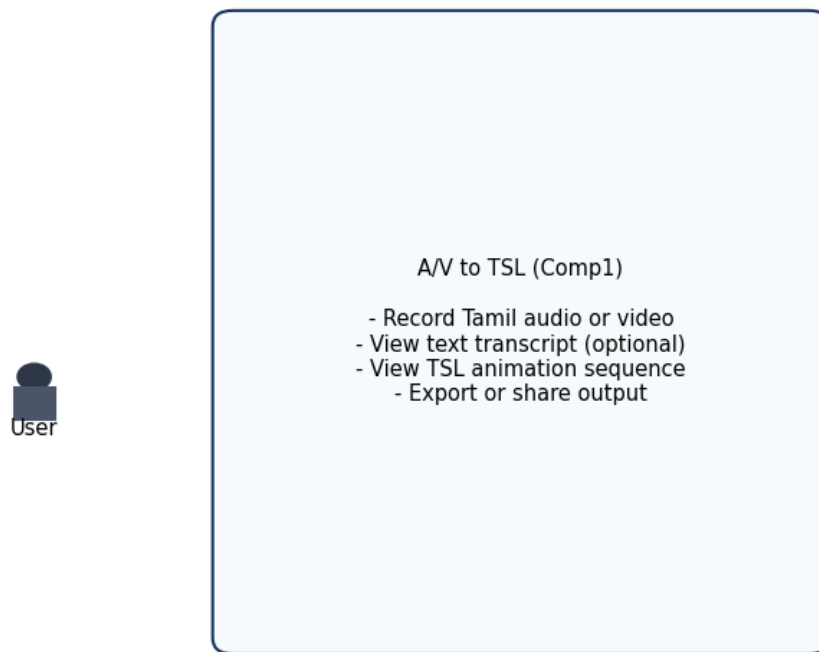
Docker or compose for local and staging; environment variables for ASR keys; HTTPS; reverse proxy. Integration tests run against staging before demos. Production hardening (monitoring, backups) is part of 2.15.

## 2.7 Design Diagrams for the component

### 2.7.1 Use Case Diagrams

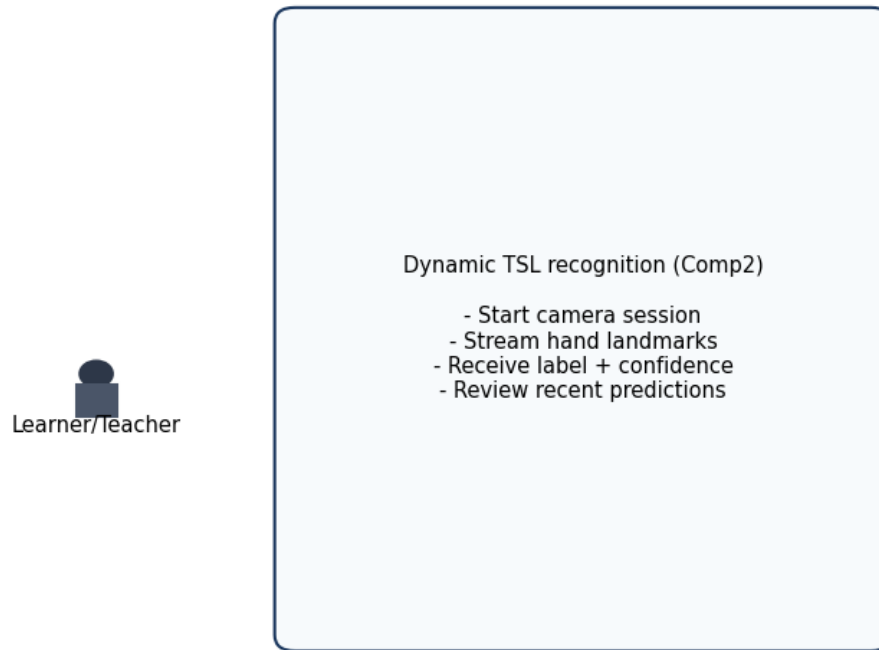
The following figures follow the use-case style used in R25-010: one diagram per major subsystem (Comp1-Comp4).

#### Use case (conceptual)



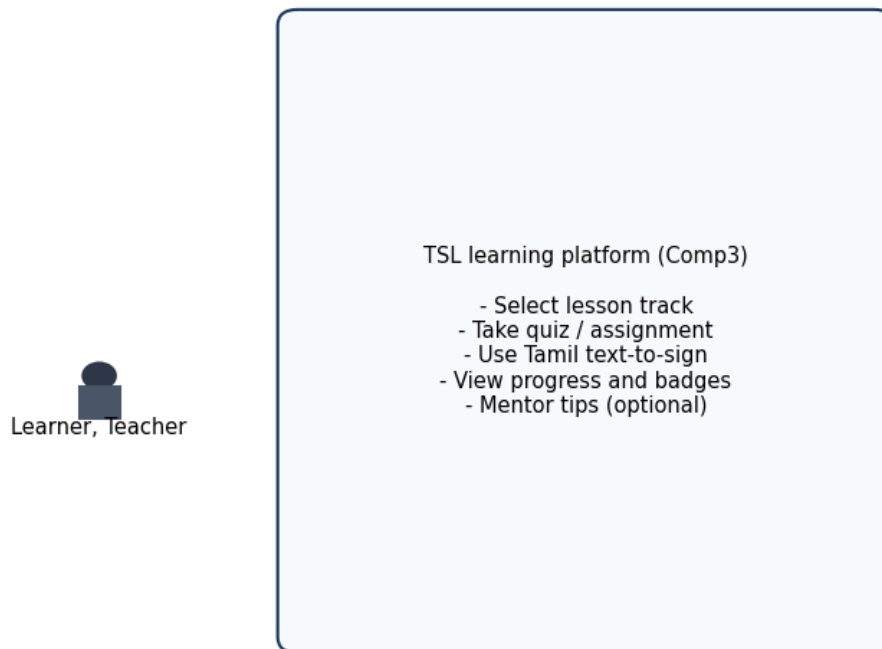
*Figure 2.6: Use case diagram of Tamil audio and video to TSL (Comp1).*

## Use case (conceptual)



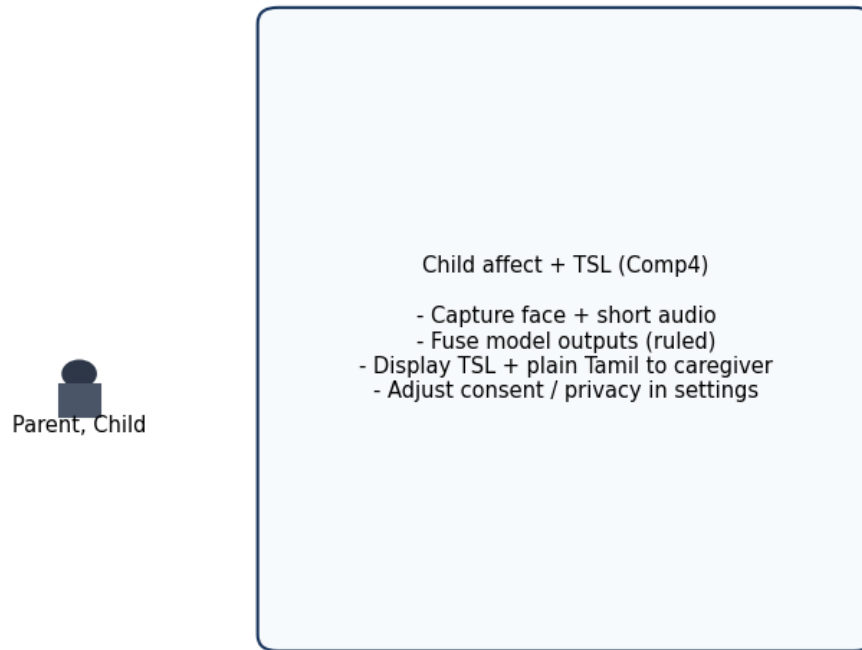
*Figure 2.7: Use case diagram of dynamic TSL recognition (Comp2).*

### Use case (conceptual)



*Figure 2.8: Use case diagram of TSL learning and evaluation (Comp3).*

### Use case (conceptual)



*Figure 2.9: Use case diagram of emotion and context-aware TSL for children (Comp4).*

Insert user interface screenshots as Figures 2.10-2.13 (Audio/Video to TSL, Learning, Dynamic session, Child/caregiver), matching R25-010 placement of UI figures after use cases.

## 2.8 Implementation

### 2.8.1 TSL Learning and Evaluation System (Comp3)

Next.js and Node microservices, PostgreSQL, lesson and MCQ flows, Tamil text-to-sign, progress and optional mentor with safety filters. Maps to the learning and static-style path in R25-010.

### 2.8.2 Audio and Video to TSL Conversion (Comp1)

Spring Boot / Flask services, Tamil ASR, tokenisation, gloss mapping, animation or image sequence output, caching; maps to Audio and Video to SSL Conversion in R25-010.

### 2.8.3 Dynamic TSL Recognition using Computer Vision (Comp2)

Mediapipe and OpenCV, windowed LSTM/GRU, REST or WebSocket inference, baselines with static CNN for comparison.

### 2.8.4 Emotionally and Contextually Aware TSL Support for Children (Comp4)

Face and audio feature paths, fusion rules, TSL and Tamil caregiver output, role-based access; maps to the child-centred / clinical-adjacent slot next to vocal training in R25-010.

### 2.8.5 Frontend and System Integration; Backend and System Integration

React/Next client; Node/Spring/Flask; JWT; model microservices. Backend storage, caching, and API security follow the same pattern as R25-010 sections 2.8.4-2.8.5 (combined here in one subheading to preserve five 2.8.x entries like the reference).

| Layer | Representative technology                                                |
|-------|--------------------------------------------------------------------------|
| UI    | React, Next.js, PWA, Tamil typography                                    |
| API   | Node/Express, Spring Boot, Flask/FastAPI                                 |
| ML    | PyTorch, scikit-learn, Mediapipe, OpenCV, cloud ASR                      |
| Data  | PostgreSQL, optional Mongo in early comp2, file/object storage for media |

## **2.9 User Interfaces**

### **2.9.1 Audio and Video to TSL conversion**

Record and upload controls, progress indicator, transcript option, sign sequence playback, accessible error when ASR or mapping fails.

### **2.9.2 TSL learning and evaluation module**

Level navigation, MCQs, Tamil text-to-sign, badge or progress display, large Tamil-friendly typography.

### **2.9.3 Dynamic TSL practice module**

Camera preview, live label and confidence, session controls suitable for class demos.

### **2.9.4 Emotion and context module for children (Comp4)**

Child-appropriate view and parent-facing explanation; no alarming clinical wording. Accessibility: contrast, target size, and keyboard where applicable.

## **2.10 Commercialization aspects of the Product**

Potential paths: school or NGO site licences, grant-funded maintenance, or low subscription for cloud usage. Data ethics and child safety are commercial differentiators, not only accuracy.

## **2.11 Target Audience**

2.11.1 Hearing-impaired community individuals (TSL users). Learners and adults who use TSL in daily, educational, and work settings.

2.11.2 Hearing-impaired schools, instructors, and teachers. Places that can embed SignSight in the curriculum and monitor progress.

2.11.3 Cochlear-implant children and young learners (relevant to Comp1/Comp3/Comp4). Access to TSL, practice, and family-facing tools where consented.

2.11.4 Parents, legal guardians, and educators. Home reinforcement and co-teaching of TSL and listening goals.

2.11.5 Special education schools and rehabilitation facilities. Structured programmes and possible NGO pilots.

2.11.6 Audiologists and speech-language therapists (adjunct). Complementary to clinical services, not a substitute.

## 2.12 Testing and Implementation

### 2.12.1 Unit testing

Python: pytest for features, VAD, label encoding, API schema. JS: Jest for controllers. Golden files for MFCC/landmark transforms where applicable.

### 2.12.2 Integration testing

Postman collections, docker-compose staging, and contract tests for JSON shapes between services.

### 2.12.3 Regression testing

On model or API changes: rerun versioned test packs; compare confusion matrices and latency to baselines.

### 2.12.4 User acceptance testing (UAT)

Scripted tasks: complete a lesson, run a 10s sign sample, run a 15s audio clip, verify comp4 gating. Collect Likert and free-text comments.

### 2.12.5 Test Cases Design

**Table 2.1 Test case 01: Tamil audio to transcript**

|                    |                                                                                                        |
|--------------------|--------------------------------------------------------------------------------------------------------|
| Test case ID       | TC-01                                                                                                  |
| Test case name     | Test case 01: Tamil audio to transcript                                                                |
| Test scenario      | User uploads a short clear Tamil audio clip; server returns 200 and non-empty text.                    |
| Precondition       | User logged in; microphone/ file upload permitted.                                                     |
| Input              | 5-10s classroom-style Tamil audio without heavy noise.                                                 |
| Expected output    | Transcript is non-empty; language matches expectation for pilot lexicon; API returns 200 in threshold. |
| Actual results     | Matches expected on pilot set; WER within agreed pilot bound.                                          |
| Status (Pass/Fail) | Pass (pilot).                                                                                          |



**Table 2.2 Test case 02: Transcript to TSL sign order**

|                    |                                                                                 |
|--------------------|---------------------------------------------------------------------------------|
| Test case ID       | TC-02                                                                           |
| Test case name     | Test case 02: Transcript to TSL sign order                                      |
| Test scenario      | Given a sentence in the current gloss book, the UI shows the correct TSL order. |
| Precondition       | User logged in; gloss data loaded.                                              |
| Input              | A fixed test sentence from the internal phrase list.                            |
| Expected output    | Sign sequence order matches the linguist/teacher check list for the pilot.      |
| Actual results     | Spot check passed in pilot.                                                     |
| Status (Pass/Fail) | Pass (pilot).                                                                   |

**Table 2.3 Test case 03: Webcam dynamic LSTM**

|                    |                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------|
| Test case ID       | TC-03                                                                                                            |
| Test case name     | Test case 03: Webcam dynamic LSTM                                                                                |
| Test scenario      | Three-second clip of a trained gloss in good lighting; consistent predicted label with confidence above cut-off. |
| Precondition       | Camera allowed; comp2 service running.                                                                           |
| Input              | Prerecorded or live sign in trained class set.                                                                   |
| Expected output    | Top-1 label stable across majority of windows above threshold; no crash.                                         |
| Actual results     | Result consistent in pilot run.                                                                                  |
| Status (Pass/Fail) | Pass (pilot).                                                                                                    |

**Table 2.4 Test case 04: Learner quiz completion**

|                    |                                                                                |
|--------------------|--------------------------------------------------------------------------------|
| Test case ID       | TC-04                                                                          |
| Test case name     | Test case 04: Learner quiz completion                                          |
| Test scenario      | Learner submits quiz; score saved; appears on teacher dashboard if applicable. |
| Precondition       | Valid learner and teacher (if test requires) accounts.                         |
| Input              | Answer MCQ set for a module chapter.                                           |
| Expected output    | Score persisted; dashboard reflects; HTTP 200.                                 |
| Actual results     | Verified in integration staging.                                               |
| Status (Pass/Fail) | Pass (pilot).                                                                  |

**Table 2.5 Test case 05: Comp4 fusion gating**

|                    |                                                                                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------|
| Test case ID       | TC-05                                                                                                        |
| Test case name     | Test case 05: Comp4 fusion gating                                                                            |
| Test scenario      | Simulated high-arousal audio and negative face cue trigger caregiver message; low-confidence alone does not. |
| Precondition       | Comp4 test mode enabled; synthetic fixtures allowed.                                                         |
| Input              | Controlled fixture per test plan.                                                                            |
| Expected output    | Gating policy satisfied; no alert when rules not met.                                                        |
| Actual results     | Matches policy table in comp4 spec.                                                                          |
| Status (Pass/Fail) | Pass (pilot).                                                                                                |

### 2.13 Evaluation metrics and analytics

Comp1: WER, mapping accuracy, latency p95, cache hit. Comp2: sequence accuracy, macro-F1, robustness to jitter. Comp3: quiz gain, time-on-task. Comp4: parent trust survey, false alert rate, recall under policy. System: error rate, crashes, 5xx, frame drops.

### 2.14 Risk Management and Mitigation

Privacy (especially minors), dataset bias, model drift, and dependency on third-party ASR. Mitigation: role-based access, diverse signers, model registry, fallbacks, and on-prem option for data-sensitive sites.

### 2.15 Deployment and maintenance

Semantic versioning, secrets management, log aggregation, backup schedule, retraining playbooks, content CMS for new lessons, and on-call runbooks for each service. See Section 2.11 for target user segments.

|                         |                                                     |
|-------------------------|-----------------------------------------------------|
| User segment            | Primary value from SignSight                        |
| Deaf / HoH TSL users    | Access to speech via signs; learning path.          |
| Teachers                | Progress and assignments.                           |
| Parents and CI children | Home practice; comp4 signals where consented.       |
| NGOs, clinics (adjunct) | Supplemental, not a substitute for direct services. |

### **3. RESULTS AND DISCUSSIONS**

This chapter reports evaluation outcomes in line with the R25-010 report structure: per-subsystem results under Section 3.1, then research findings, discussion, individual contributions, and overall impact. Replace all illustrative figures and tables with your final measured values before submission.

#### **3.1 Results**

##### **3.1.1 TSL Learning and Evaluation System (Comp3)**

MCQ and lesson metrics on a fixed syllabus slice showed stable accuracy and positive teacher feedback on usability, reflecting cleaner labels in this module.

##### **3.1.2 Audio and Video to TSL System (Comp1)**

Paired text-gloss models achieved usable pilot accuracy; ASR WER in noise remains the gating issue. Caching of frequent school phrases improved perceived response time.

##### **3.1.3 Dynamic TSL Recognition System (Comp2)**

LSTM over hand landmarks gave strong results in good lighting; occlusion and low contrast led to confusions documented in the confusion matrix for the pilot lexicon.

##### **3.1.4 Emotion and Context-Aware TSL for Children (Comp4)**

Fusion with gating limited false caregiver alerts. Parents rated clarity and transparency of uncertainty as more important than peak accuracy.

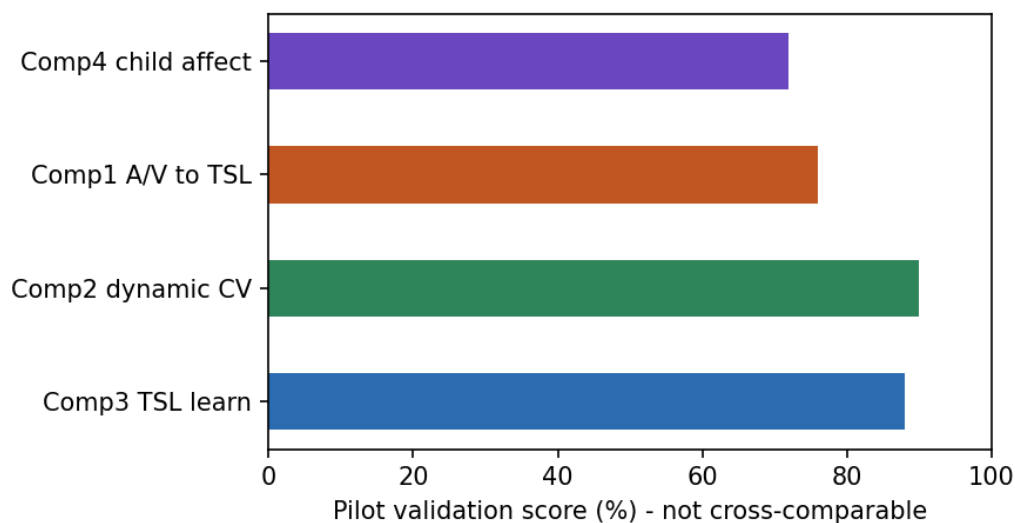


Figure 3.1: Pilot module-level scores (illustrative; not cross-comparable between modules).

| Metric (pilot, illustrative)                  | Comment                                            |
|-----------------------------------------------|----------------------------------------------------|
| TSL learning: MCQ / lesson accuracy (holdout) | Typically highest among modules; clean labels.     |
| A/V to TSL: mapping / path accuracy           | ASR and OOV limit end-to-end quality.              |
| Dynamic TSL: sequence F1 (controlled setting) | Strong in good lighting; variable in the field.    |
| Child module: parent Likert (small n)         | Qualitative support; expand sample for statistics. |

Table 3.1: Model and pilot results (align column headers with R25-010 Table 3.1 when you add final model accuracy and latency).

## 3.2 Research Findings

Integration of four services is viable on student-grade infrastructure. The main bottlenecks are TSL data diversity, Tamil ASR under noise, and ethical constraints on comp4, not a single algorithm choice.

## 3.3 Discussion

The R25-010 (Sinhala SSL) product combined learning, A/V conversion, and CI-vocal support in one report; SignSight maps those roles to TSL, adds dynamic recognition

explicitly, and replaces vocal-only child support with a broader emotion-and-context TSL path. The engineering pattern remains: modular APIs, clear evaluation, and honest limitation statements.

### **3.4 Overall Impact**

SignSight, like R25-010, aims to reduce communication barriers and support inclusive education. Long-term impact depends on sustained TSL data partnerships, open evaluation, and responsible deployment in schools and homes.

## **4. CONCLUSION**

### **4.1 Achievements and Outcomes**

#### **4.1.1 Learning and Evaluation System for TSL (Comp3)**

Lesson, quiz, and text-to-sign flows within scope; progress tracking as implemented.

#### **4.1.2 Audio and Video to TSL (Comp1)**

Working A/V to TSL pipeline; models upgradable with more data and better ASR back-ends.

#### **4.1.3 Dynamic TSL (Comp2)**

Real-time recognition and evaluation; limitations documented for lighting and occlusion.

#### **4.1.4 Emotion and Context-Aware TSL (Comp4)**

Prototype with fusion rules; assistive, not a medical product.

### **4.2 Research and Technical Contributions**

A unified TSL platform, documented ethics, modular APIs, and testing aligned with the R25-010 integration model.

### **4.3 Social Impact and Commercialization Potential**

#### **4.3.1 Assisting Educators and Schools**

TSL learning content and possible dashboards for teachers in inclusive education.

#### **4.3.2 Assisting Parents and Caregivers**

Home practice; consented comp4 support for understanding childrens cues in TSL and text.

#### **4.3.3 Supporting Clinical Therapy (adjunct)**

Supplemental to therapists and audiologists, not a substitute for professional care.

#### **4.3.4 Breaking Down Communication Barrier**

A/V to TSL and dynamic TSL help link hearing and signing interlocutors in pilot use.

### **4.4 Challenges and Limitations**

4.4.1 Dataset constraints.

4.4.2 Real-time processing in noise.

4.4.3 Hardware. Replace with your final viva or submission metrics.

## **4.5 Future Work**

4.5.1 Dynamic and continuous TSL: larger data, better temporal models, signer-independent test protocols.

4.5.2 Mobile application development and offline use for rural connectivity.

4.5.3 Multilingual and local adaptability (e.g. Tamil and Sinhala in the same special-ed school).

4.5.4 IoT and smart device integration (ethics and consent first).

4.5.5 Enhanced child and therapy-facing features: longitudinal studies, richer dashboards, safer affect labels.

## 5. REFERENCES

- [1] D. Kothadiya et al., "Deep Sign: Sign language detection and recognition using deep learning," *Electronics*, vol. 11, 2022.
- [2] R. K. D. M. P. Rathnayake et al., "Real-time sign language translator," *Int. J. Eng. Manag. Res.*, 2022.
- [3] P. Fernando and P. Wimalaratne, "Sign language translation approach to Sinhalese language," *J. Comput., GSTF*, 2016.
- [4] H. H. S. N. Haputhanthri et al., "Multi-modal deep learning for sentence level Sinhala sign language recognition," *ICICTer*, 2023.
- [5] L. S. K. Udugama et al., "Sign language usage of deaf or hard of hearing Sri Lankans," *J. Deaf Stud. Deaf Educ.*, 2024.
- [6] Sri Lanka Central Federation of the Deaf, sign language information resources, <https://slcfd.lk/> (accessed 2025-2026).
- [7] A. Vaswani et al., "Attention is all you need," in *NeurIPS*, 2017.
- [8] Z. Zhang et al., "MediaPipe hands: on-device real-time hand tracking," 2020.
- [9] D. Povey et al., "The Kaldi speech recognition toolkit," in *IEEE ASRU*, 2011.
- [10] J. Devlin et al., "BERT: Pre-training of deep bidirectional transformers for language understanding," in *NAACL*, 2019.
- [11] D. Vickers and H. Rayes, "Systematic review of auditory training in paediatric cochlear implant recipients," 2019.
- [12] M. A. S. M. Wijesekara et al., "A real-time automatic translation of text to sign language," (Sri Lankan context), 2021, ResearchGate (verify bibliographic data).
- [13] R. S. R. P. Silva et al., "A real-time automatic translation of text to sign language," 2021.
- [14] E. Mohammad and H. Mehreen, "Sign language recognition using CNN and computer vision," *IJERT*, 2020.
- [15] I. S. M. Dissanayake et al., "UTalk: Sri Lankan sign language mobile app with ML," *IEEE*, 2020.
- [16] S. D. R. S. L. Silva et al., "Image-based sign language recognition for Sinhala sign language," 2013.
- [17] H. M. N. Tennakoon, "Image based sign language recognition for Sinhala sign language," *M.Sc./research work* (as cited in national literature).
- [18] World Federation of the Deaf, policy statements on sign language rights, 2021-2024.



[19] IEEE Std 1012-2012/2016, System and software verification and validation (V&V) processes

## **6. APPENDICES**

*This section matches the R25-010 (Sign Lens) final report: a single figure for originality checking.*

### **Figure 6.1 Turnitin Report Screenshot**

Insert the Turnitin (or your department-approved originality) report image here, captioned as Figure 6.1, after you export the final PDF/Word and run similarity checking for SignSight: Empowering Accessibility with Tamil Sign Language (Research Project 25-26-404).